DBCFS-Net: Dynamic Blocks and Cross-Domain Feature Synergy for Remote Sensing Object Detection

Qingyao Lin, Yuhui Zheng, *Member, IEEE*, Le Sun, *Senior Member, IEEE*, Shihao Dong, Rugang Wang, and Feng Zhou

***Abstract*—The detection of objects in remote sensing images represents a significant subtask in the domain of computer vision. However, it poses some challenges: mitigating high-frequency fine-grained feature loss, capturing efficient and discriminative object features, and improving separability between complex backgrounds and objects. Therefore, we propose a dynamic blocks and cross-domain feature synergy (DBCFS-Net) for remote sensing object detection. First, we propose a dynamic block downsampling module (DBDM). Its block partitioning strategy and dynamic offsets overcome the limitations of rigid downsampling that cause blurred features, while adaptively focusing on critical target regions to preserve high-frequency fine-grained details. Second, a cross-domain aware feature enhancement module (CDAFEM) is introduced to enhance the model's adaptability to scale variations and complex backgrounds by establishing consistency constraints through cross-domain joint modeling, thereby preserving discriminative feature information. Concurrently, it refines object boundaries and edge structures through texture information optimization, reinforcing edge continuity to improve the model's generalization capability and robustness. Finally, a mamba-based local–global feature co-promotion module (MB-LGFCPM) is designed to guide the model to couple local details with global contextual information, constructing a bidirectional 'detail-semantic' enhancement mechanism for feature representation. This achieves the dynamic interaction and synergistic complementarity optimization between features. Extensive experimental results validate the effectiveness of the proposed method, achieving mAP scores of 87.8%, 96.9%, and 96.6% on DIOR, RSOD, and NWPU VHR-10 datasets—surpassing state-of-the-art detectors.**

***Index Terms*—Cross-domain feature augmentation, feature dependency, multiscale dynamic feature fusion, object detection, remote sensing images.**

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I. Introduction

With the advancement of optical remote sensing technology, the object detection in remote sensing imagery has emerged as an active research area. This task aims to identify both the categories and spatial distributions of specific objects within remote sensing images. As a critical component of remote sensing image analysis, it plays a crucial role in domains including military reconnaissance [1], environmental conservation [2], agricultural monitoring [3], and traffic management [4].

Nevertheless, compared with conventional scenarios, remote sensing object detection remains challenging due to complex scene environments, diverse object scales, and feature interference among densely distributed objects, as illustrated in Fig. 1. Remote sensing images are typically acquired from large field-of-view (FOV) observations, where objects occupy only a minimal proportion of pixels in comparison with the expansive and cluttered backgrounds [5], [6]. Also, persistent challenges in remote sensing object detection arise from insufficient feature representation due to low-resolution objects and poor object–background contrast. Concurrently, the distinctive attributes inherent to remote sensing imagery remain underexplored in optimizing feature extraction performance [7]. In terms of downsampling, the currently widely used fixed-domain downsampling method cannot adapt to the geometric shape of the target and is accompanied by irreversible loss of spatial details. Fixed windows rigidly cut off the boundaries of the target or destroy the structured information of irregular targets, resulting in blurred boundaries and shape distortion, which further exacerbates the loss of features of small targets in remote sensing images.

Researchers have undertaken efforts from multiple perspectives to address these challenges. Data augmentation techniques have been employed to enhance the feature robustness against scale-induced feature mismatch [8], [9] [10]. Also, advances in deep learning methodologies offer promising directions for tackling these challenges. Liu et al. [11] introduced ABNet, which adaptively integrates channelwise and spatial-location information to acquire more diverse discriminative features. This approach addresses scale variation challenges by enhancing semantic richness.

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Fig. 1. Remote sensing imagery exhibits the following distinctive characteristics. (a) Scene complexity: manifested by the presence of diverse object types within a single scene, which can introduce significant interference for object identification. (b) Diversity in object scales: characterized by co-existing objects exhibiting substantial variations in size and arbitrary orientations within the same scene. (c) Object density: evidenced by closely spaced objects within a scene, often resulting in blurred interobject boundaries.

Zhang et al. [12] introduced CoF-Net, a progressive network that incrementally refines primitive features toward discriminative representations characterized by rich spectral details and semantic dependencies. Ma et al. [13] proposed AFSNet, which decomposes object features into distinct channels to effectively circumvent interfeature interference and enhance detection efficacy. Gao et al. [14] developed AGMF-Net, which reinforces multiscale feature representations by enhancing high-level semantic information and capturing salient global features while mitigating background interference.Tong et al. enhanced small object detection performance by expanding receptive field ranges and integrating spatial detail features with contextual semantic information [15]. Li et al. [16] proposed TBNet, which enhances the discriminability of textural features by exploring interpixel relationships to accentuate discrepancies between objects and backgrounds in remote sensing imagery. Although the aforementioned methods have achieved notable progress from diverse perspectives, they overlook the significance of establishing long-range dependencies through feature affinity and coupling local–global characteristics. These mechanisms play crucial roles in distinguishing objects of interest from image backgrounds. Furthermore, how to efficiently retain discriminative favorable features and mitigate high-frequency irreversible information loss are also issues worthy of further in-depth investigation.

In this article, a method is proposed to mitigate feature loss and enhance feature perception through dynamic block downsampling and cross-domain feature synergy, addressing the challenges of complex background interference and suboptimal feature extraction in remote sensing imagery. The principal contributions are summarized as follows.

1) A novel method called dynamic blocks and cross-domain feature synergy (DBCFS-Net) is proposed for the object detection in remote sensing images. The method forms a step-by-step, functionally coupled feature optimization link by constructing a synergistic mechanism of "feature retention-cross-domain enhancement-complementary fusion," and provides an effective solution for remote sensing target detection in complex scenarios by utilizing this end-to-end synergistic design. Comprehensive evaluations were conducted on multiple benchmark datasets (DIOR, RSOD, and NWPU VHR-10). Experimental results demonstrate that the proposed DBCFS-Net method achieves robust detection performance across all three datasets, attaining SOTA results.
2) The DBDM is introduced, which not only effectively mitigates feature loss during downsampling but also addresses the inherent limitations of rigid modeling in conventional fixed-grid approaches, specifically their neglect of spatial heterogeneity and geometric transformations in objects.
3) The CDAFEM is proposed, in which a tri-stage optimization mechanism for cross-domain synergy is constructed through channel recalibration, spatial focusing, and frequency-domain enhancement. This approach preserves discriminative information while suppressing redundant noise. Concurrently, an edge-aware structure is designed to perform explicit edge modeling by leveraging textural features of objects, thereby mitigating indistinct object boundaries and enhancing foreground–background separability.
4) The MB-LGFCPM is proposed to enhance long-distance positional interactions for global modeling while preserving local detail features, thereby achieving complementary information fusion.

The remaining structure of this article is organized as follows. Section II provides a concise review of related works. Section III presents the architecture of the proposed DBCFS-Net methodology, followed by detailed elaborations on the structures of the DBDM, CDAFEM, and MB-LGFCPM. Experimental results and corresponding analyses are given in Section IV. Concluding remarks are provided in Section V.

II. Relate Works

A. Deep Learning-Based Remote Sensing Image Detection

Unlike traditional methods reliant on handcrafted features such as shape, texture, and fixed scales [17], [18], [19], the advent of deep learning has enabled convolutional neural network (CNN)-based object detectors to adaptively extract image features. These approaches preserve richer semantic characteristics while exhibiting enhanced robustness. Currently, deep learning-based object detectors are mainly divided into two-stage detectors [20], [21], single-stage detectors [22], [23], and Transformer-based detectors [24], [25]. These methods have driven the development and advancement of object detection technology.

The application of these object detectors to remote sensing imagery has led to significant progress, primarily through architectural adaptations that address domain-specific characteristics. For instance, Li et al. [26] developed an adaptive points assessment and assignment model for rotated object detection, which demonstrated superior performance but was limited by feature-instance misalignment when processing oriented bounding boxes. Shao et al. [27] reduced interclass feature interference by leveraging interclass similarity through a cross-entropy loss, though the method incurred high computational costs due to extensive concatenation operations.

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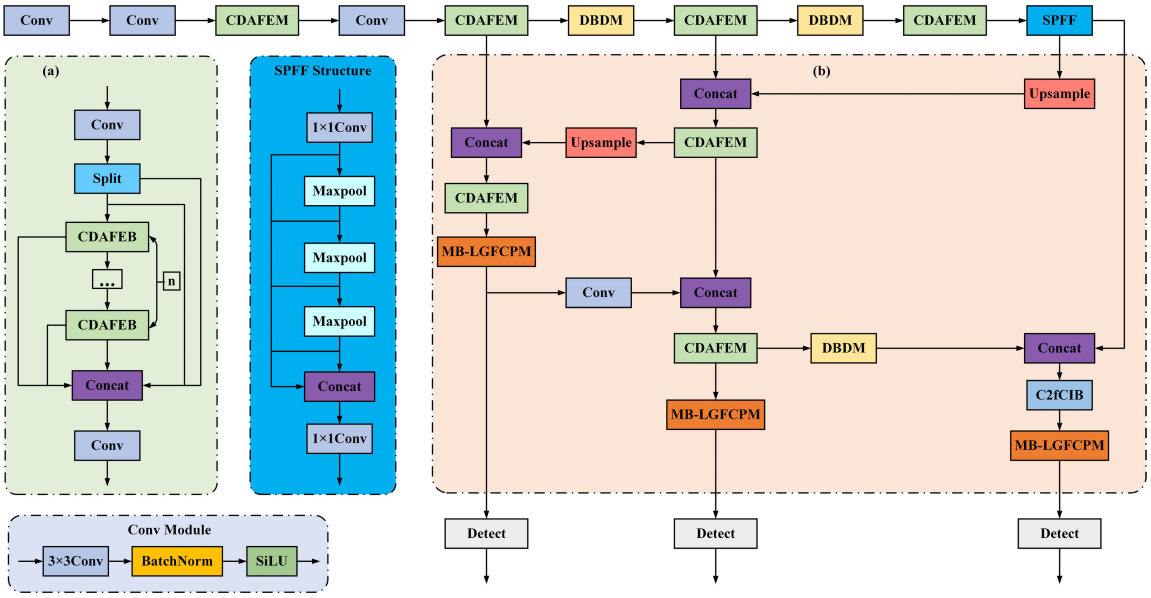


Fig. 2. Overall architecture of the proposed DBCFS-Net method. The backbone network consists of a feature extraction structure coupled with a spatial pooling feature pyramid. Within the feature extraction component, each group beyond the first is composed of a convolutional block and a CDAFEM. (a) CDAFEM adopts a residual architecture comprising one convolutional block and n stacked CDAFEB. (b) MB-LGFCPMP constitutes an enhanced pyramid structure designed to capture local–global features through the MB-LGFCPM framework, as detailed in Section III-D. Finally, detection is performed by three distinct heads optimized for objects at varying scales.

Yue et al. [28] proposed SCFNet, which refines semantic features by computing global-local feature similarity, thereby enhancing detection accuracy. Nevertheless, the approach still exhibits suboptimal performance in detecting small objects in remote sensing imagery. Ren et al. [29] proposed a single-stage remote sensing object detection method called StrMCsDet based on single-layer features, which effectively reduced computational overhead but did not fully consider the issue of large differences in remote sensing object sizes. The YOLO series, as prominent single-stage detectors, have also been extensively adapted for remote sensing. The pioneering work YOLT facilitated multiscale object detection through dual-scale image processing [30]. Subsequent improvements were also numerous, including Zhu et al. [31] incorporated Transformer modules into the backbone to capture global contexts and long-range dependencies. Wang et al. [32] employed deformable convolutions in the neck network to enhance feature fusion and bridge semantic gaps. While Zhu et al. [31] and Wang et al. [32] effective, such structural additions often lead to increased parameter counts. To address this, lightweight attention mechanisms, such as the coordinate attention module, have been embedded in shallow networks to suppress background features and strengthen object representations by Shen et al. [33]. Alternatively, Zhang et al. [34] expanded dilated convolutions to expand the receptive field and guided feature dependency by constructing global contextual relationships.

B. Feature Enhancement and Fusion Methods

The limited spatial extent of objects in remote sensing imagery poses a fundamental challenge for discriminative feature extraction. To overcome this, diverse methodological frameworks have been explored. Data augmentation strategies, such as the reinforcement learning-based policy search by Cubuk et al. [35] and the CutMix method by Pan et al. [36],

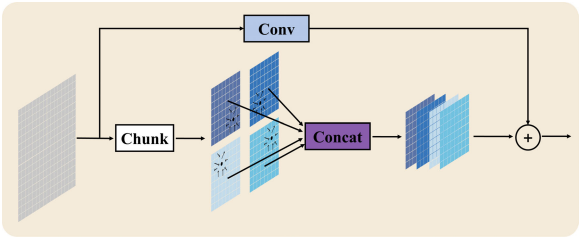


Fig. 3. Dynamic block downsampling module. By integrating a dynamic offset mechanism with a feature partitioning strategy, our approach focuses on critical object regions while preserving fine-grained spatial information (e.g., edge structures and textural patterns). This design imbues the model with intrinsic adaptability to geometric transformations while simultaneously preventing cross-partition feature contamination, thereby enhancing training stability and feature discriminability.

have been employed to the enhance feature robustness and semantic diversity. Beyond data-centric strategies, the design of efficient feature extraction and fusion architectures is equally critical. The feature pyramid network (FPN) is a foundational architecture that constructs multiscale representations by combining high-resolution details with rich semantics [37]. However, FPN relies heavily on low-resolution semantic pathways, which can degrade spatial fidelity. To address this, Li et al. [38] introduced a location refinement mechanism that explicitly integrates low-level positional cues, enhancing feature representations within object regions, though boundary semantics for irregular objects remain under-optimized. Wang et al. [39] have explored multibranch fusion with bridging strategies to facilitate cross-scale interactions and reduce semantic gaps. A limitation of such direct fusion is the potential suppression of object information by background noise. Consequently, scale-equilibration mechanisms have been proposed by Wang et al. [40] to dynamically recalibrate multiscale feature contributions, preventing dominance by any single scale. More recently, Chen et al. [41] incorporated Transformer

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modules into the feature pyramids to model global contexts and establish long-range dependencies, significantly improving detection, especially for dispersed objects.

III. Proposed Method

A. Overview

The overall architecture of the proposed method DBCFS-Net is illustrated in Fig. 2. This framework comprises three core components: DBDM, CDAFEM, and MB-LGFCPM. Comprehensive technical specifications for these modules are detailed in Sections III-B–III-D.

B. Dynamic Block Downsampling Module

Fixed-kernel downsampling in remote sensing object detection often causes irreversible loss of spatial details and blurred object boundaries, which severely limits the detection performance. To mitigate this issue, we propose a DBDM, as illustrated in Fig. 3, which effectively preserves structural information while remaining computationally efficient. The module first partitions the input feature map into four regions along the horizontal and vertical axes for separate processing. This strategy helps retain fine-grained details within local blocks. A dynamic sampling grid, derived from normalized coordinates and content-adaptive offsets, is then applied to enable geometry-aware feature extraction that adapts to object shapes. The independently processed regional features are subsequently concatenated and fused. In addition, a residual connection is introduced to facilitate gradient flow and strengthen feature representation. The operational workflow proceeds as follows:

$$\text{Offset} = \text{Conv}\left(\text{ReLU}\left(\text{Conv}\left(x\right)\right)\right) \tag{1}$$

$$\text{Block}_k = \text{Chunk}\left(x, g_{\text{base}} + \text{Offset}_k\right) \tag{2}$$

$$\text{Feat}_k = \text{Conv}\left(\text{Block}_k\right) \tag{3}$$

$$\text{Fusefeat} = \text{Cat}_{k=1}^{N}\left(\text{Feat}_k\right) \tag{4}$$

$$\text{Output} = \text{Conv}\left(\text{Fusefeat} + \text{Conv}\left(x\right)\right). \tag{5}$$

Within this formula, Offset_k denotes the dynamic offset of the kth block, g_{base} represents the normalized coordinate grid, Block_k corresponds to the sampling result of the kth block, Chunk signifies the dynamic block sampling procedure, Fusefeat designates the features obtained through fusion after sampling, and Output indicates the output features from the DBDM.

C. Cross-Domain Aware Feature Enhancement Module (CDAFEM)

The efficacy of feature modeling is inherently limited by representational gaps across domains. In remote sensing object detection, texture is pivotal; its insufficient exploitation results in weak boundary definition and constrains the model's ability to discern structural differences. To address this limitation, we propose a CDAFEM, with its architectural implementation detailed in Fig. 4. For spatial processing, the module applies depthwise separable convolution independently extract spatial features within each channel, then fuse them cross-channel via pointwise convolutions to reduce redundancy and enhance local feature representation. The introduction of variable dilation rates adaptively adjusts receptive fields to capture multiscale context. The resultant features are then additively merged with the original input through a residual connection. This architecture not only retains pristine feature integrity but also dynamically processes multigranular spatial information, ultimately strengthening contextual continuity and augmenting feature representational capacity. The operational workflow proceeds as follows:

$$S_1 = \text{PointConv}\left(\text{ReLU}\left(\text{DWConv}_{d=1}\left(\text{Conv}\left(x\right)\right)\right)\right) \tag{6}$$

$$S_2 = \text{PointConv}\left(\text{ReLU}\left(\text{DWConv}_{d=2}\left(\text{Conv}\left(x\right)\right)\right)\right) \tag{7}$$

$$S_{\text{out}} = \sigma\left(\text{Conv}\left(\text{Concat}\left(S_1, S_2\right)\right)\right) + \text{Conv}\left(x\right) \tag{8}$$

where PointConv denotes the pointwise convolution operation, ReLU represents the rectified linear unit activation function, DWConv signifies the depthwise convolution with dilation rate d, Conv indicates the convolution operator, Concat corresponds to the feature fusion process, σ designates the sigmoid activation function. S_1 and S_2 refer to feature maps processed through distinct branches, and S_{out} represents the final output spatial domain feature map.

For channel-domain processing, the input features are split into two parallel branches through convolution. The first branch employs gated convolutional layers that dynamically recalibrate channelwise contributions based on feature activation significance, enabling input-adaptive channel selection. The second branch constructs a cross-channel covariance matrix to explicitly capture interfeature dependencies. The gated branch output is further compressed via global average pooling and integrated with the covariance matrix through matrix multiplication. This dual-branch design simultaneously preserves original information and strengthens feature interactions, substantially amplifying the model's representational capacity. The complete operational workflow proceeds as follows:

$$C_1 = \text{PartConv}\left(\text{Conv}\left(x\right)\right) + \text{Conv}\left(x\right) \tag{9}$$

$$C_2 = \text{Re}\left(\text{Conv}\left(x\right)\right) \otimes \text{Re}^{T}\left(\text{Conv}\left(x\right)\right) \otimes \text{GAP}\left(C_1\right) \tag{10}$$

$$C_{\text{out}} = C_1 + C_2. \tag{11}$$

In the above formula, PartConv denotes gated partial convolution, Re represents the reshape operation, T signifies the matrix transpose operator, $\otimes$ indicates element-wise multiplication, GAP corresponds to global average pooling. C_1 and C_2 refer to feature maps processed through complementary branches, and C_{out} designates the final output channel-domain feature map.

Prior to frequency-domain processing, features processed through the spatial and channel domains undergo channelwise concatenation. This integration achieves synergistic optimization of spatial minutiae and channelwise semantics, unifying geometric priors and semantic priors within a consolidated feature representation framework. Subsequent application of the Harr wavelet transform decomposes features into low-frequency components encoding global structural semantics and high-frequency subbands capturing localized edge and

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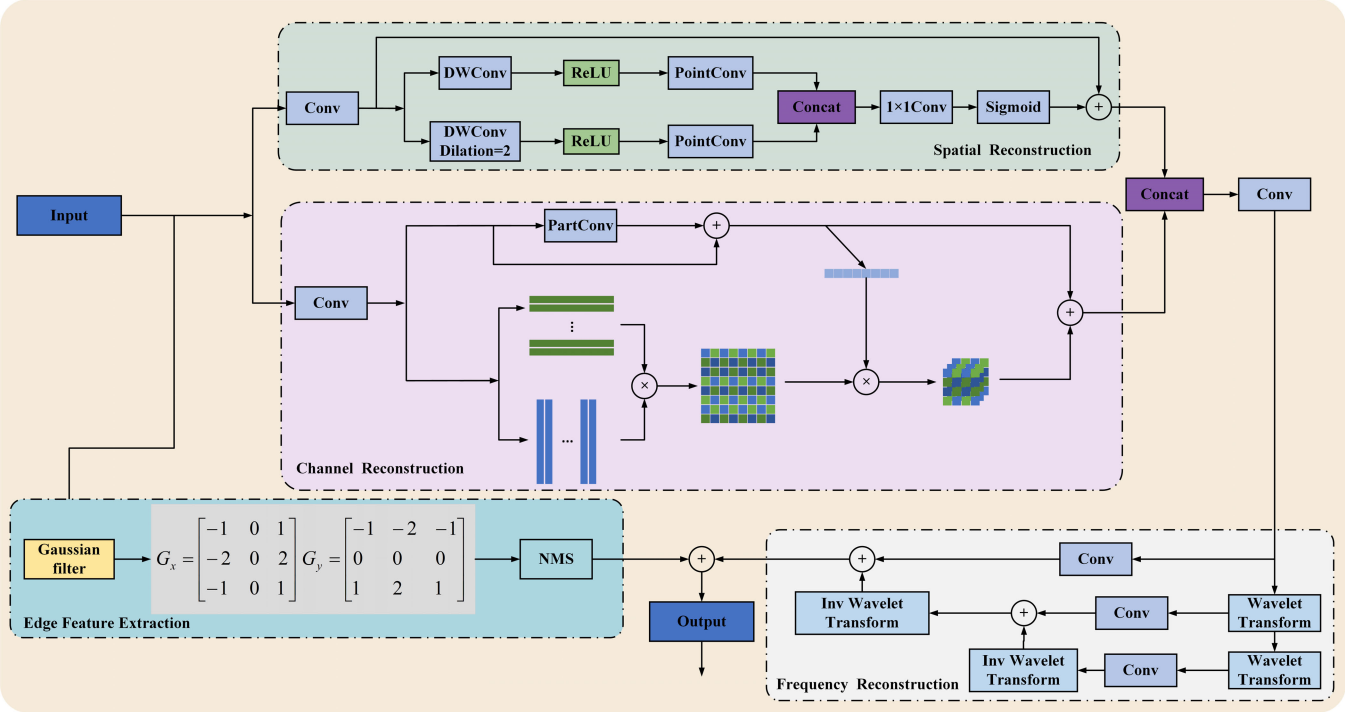


Fig. 4. Proposed CDAFEM explores the establishment of a cross-domain feature coordination representation mechanism from the perspectives of spatial domain, channel domain, and frequency domain.

texture details. This multiresolution decomposition leverages the localization properties of wavelet bases to extract discriminative frequency-domain features. Consequently, the model gains enhanced robustness to scale variations, noise corruption, and fine-grained detail preservation. Those critical capabilities for aerial object detection in complex environments. The operational workflow proceeds as follows:

$$\text{Fuse}_{s+c} = \text{Conv}\left(\text{Concat}(S_{\text{out}}, C_{\text{out}})\right) \tag{12}$$

$$F_{\text{out}} = \text{WT}\left(\text{Fuse}_{s+c}\right). \tag{13}$$

In the above formula, WT denotes the wavelet transform operation, Fuse_{s+c} epresents the cross-domain fusion of spatial and channel features, and Fuse_{out} designates the final output frequency-domain feature map.

In addition, to further reduce background and noise interference, improve the integrity of small object boundaries, and enhance texture features, we use Gaussian blurring to smooth noise while preserving primary edge structures. We then apply the Sobel operator to obtain gradients in both horizontal and vertical directions, calculating gradient magnitude and direction to determine edges. Subsequently, nonmaximum suppression is applied to further remove nonedge regions, refine edge structures, and connect broken edge structures to enhance edge continuity. Finally, the feature map obtained after the wavelet transform is added to the feature map obtained after edge extraction to produce the final feature map output. The operational workflow proceeds as follows:

$$\tilde{G} = \text{NMS}\left(\text{Conv}_{\sqrt{G_x^2+G_y^2}}(x)\right) \tag{14}$$

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \tag{15}$$

$$Y = \tilde{G} + F_{\text{out}}. \tag{16}$$

In this context, $\tilde{G}$ denotes the edge feature map obtained, NMS represents nonmaximum suppression, G_x and G_x designate the gradients in the horizontal and vertical directions, respectively. Also, Y represents the final feature map output by the cross-domain perception feature enhancement module.

D. Mamba-Based Local–Global Feature Co-Promotion Module (MB-LGFCPM)

For RSOD, precise localization and discrimination of objects across diverse scales and spatial contexts are paramount. To address this, we design MB-LGFCPM, with its architectural implementation depicted in Fig. 5. This module establishes a complementary interplay between localized details and global contextual semantics, ensuring integrity-preserving feature propagation to subsequent processing stages. Within the MB-LGFCPM, the primary goals are to enhance local–global feature fusion, improve contextual modeling, and preserve fine-grained details alongside global semantics. This integration achieves bidirectional complementarity, enriching representations and mitigating contextual ambiguities, thereby improving detection robustness in complex remote sensing scenarios.

For local detail extraction, convolution and pooling capture object-level features, which are mapped to a 1-D space to model nonlinear relationships. A gating mechanism adaptively emphasizes relevant channels and recalibrates the attention map, enabling second-order feature interactions with low computational cost and improving geometric detail representation. Position encoding further introduces local bias and spatial awareness, enhancing sensitivity to input features. In addition,

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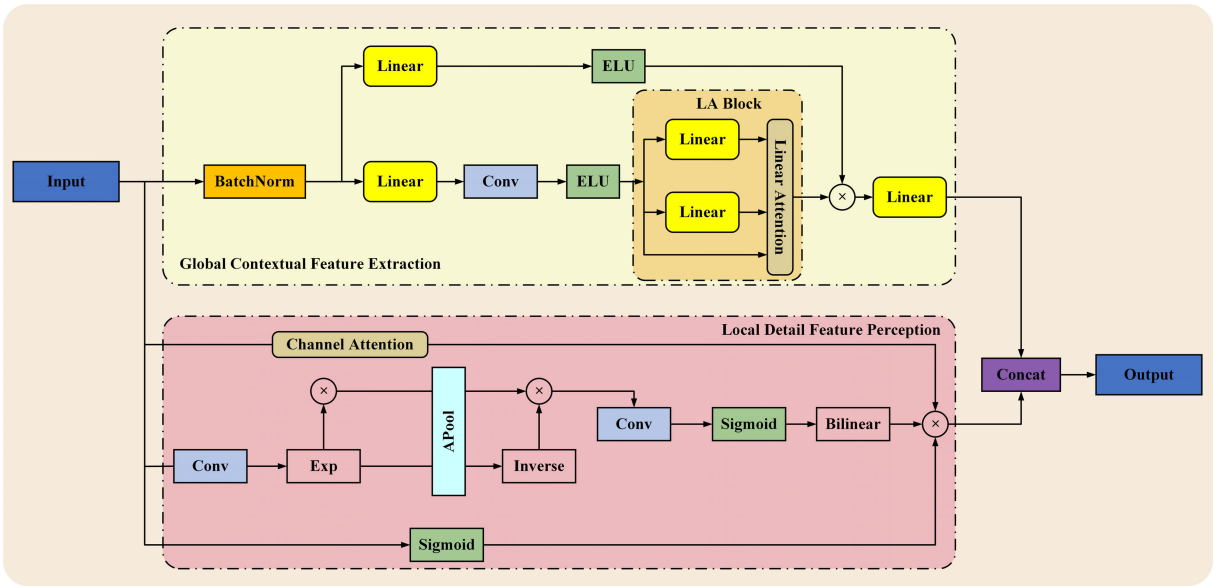


Fig. 5. MB-LGFCPM.

lightweight operations such as convolution and bilinear interpolation maintain computational efficiency. The operational workflow proceeds as follows:

$$X_{\text{ca}} = x \otimes \text{Conv1D}\left(\text{AP}\left(x\right)\right) \tag{17}$$

$$\text{Local}_i = X_{\text{ca}} \otimes \text{Ga}\left(X_{\text{ca}}\right) \otimes \ \text{Interpolate}\left(x\right). \tag{18}$$

In the aforementioned architecture, AP denotes the adaptive pooling operation, ConvlD represents the 1-D convolutional operation that projects features onto a 1-D space, X_{ca} signifies the feature map obtained after channel attention processing, Ga corresponds to the gating mechanism incorporating positional encodings, Interpolate operation employing bilinear interpolation, and Local_i indicates the final locally enhanced feature representation extracted by the local detail refinement module.

For global information extraction, we adopt a dual-path architecture that integrates linear processing with the Mamba structure. This design efficiently captures long-range dependencies and global context while suppressing local ambiguities and noise. In the fusion stage, global context features are concatenated with local detail features, enabling complementary integration across scales. This strategy strengthens feature synergy, thereby improving model generalization and robustness. The operational workflow proceeds as follows:

$$\text{Branch}_1 = \text{ELU}\left(\text{Linear}\left(\text{BN}\left(x\right)\right)\right) \tag{19}$$

$$\text{Branch}_2 = \text{LA}\left(\text{ELU}\left(\text{Conv}\left(\text{Linear}\left(\text{BN}\left(x\right)\right)\right)\right)\right) \tag{20}$$

$$\text{Global}_i = \text{Linear}\left(\text{Branch}_1 \otimes \text{Branch}_2\right) \tag{21}$$

$$\text{Fusion} = \text{Catcat}\left(\text{Local}_i, \text{Global}_i\right) \tag{22}$$

In the above formula, ELU denotes the exponential linear unit activation function, represents the linear function operation, BN signifies the batch normalization layer, LA corresponds to the linear attention mechanism, Global_i designates the final contextually aggregated feature map obtained from the global context extraction module, and Fusion indicates the integrated feature map resulting from the fusion of local detail features and global contextual features.

TABLE I
EXPERIMENTAL PARAMETER AND DATA AUGMENTATION SETTING

Parameter Name	Parameter Value	Data Augmentation	Value
Momentum	0.937	Translate	0.13
Learning-rate	0.01	Scale	0.08
Weight-decay	0.0005	Fliplr	0.09
Epoch	500	Mosaic	0.43
Batch-size	8	Others	Default

IV. EXPERIMENTS

A. Experimental Setup

During the experiment, the experimental equipment used was an NVIDIA GeForce RTX3090 24-GB GPU. Python 3.8.19 was used as the compiler, and the code was written under the PyTorch 2.4.1 framework. During model training, stochastic gradient descent (SGD) was used as the optimizer. To ensure fairness and comparability of the experiments, we strictly adhered to the widely adopted dataset partitioning method in the field, with a ratio of 7:2:1 for the training, validation, and test sets. To mitigate the category imbalance, we used a correlation data augmentation method, as shown in Table I. The method and parameters are as follows: translate set to 0.13, scale set to 0.08, fliplr set to 0.09, and mosaic set to 0.43. Other parameters are kept as the default. When training the model, we set the experimental epoch to 500 to allow the model to fully converge, and we set an early-stopping strategy to stop training if there is no improvement after waiting for 50 epochs.

B. Evaluation Metrics

To evaluate the performance of our approach, we adopt average precision (AP) and mean AP (mAP) as primary metrics for assessing detection accuracy. AP quantifies the precision–recall tradeoff for individual object categories, while

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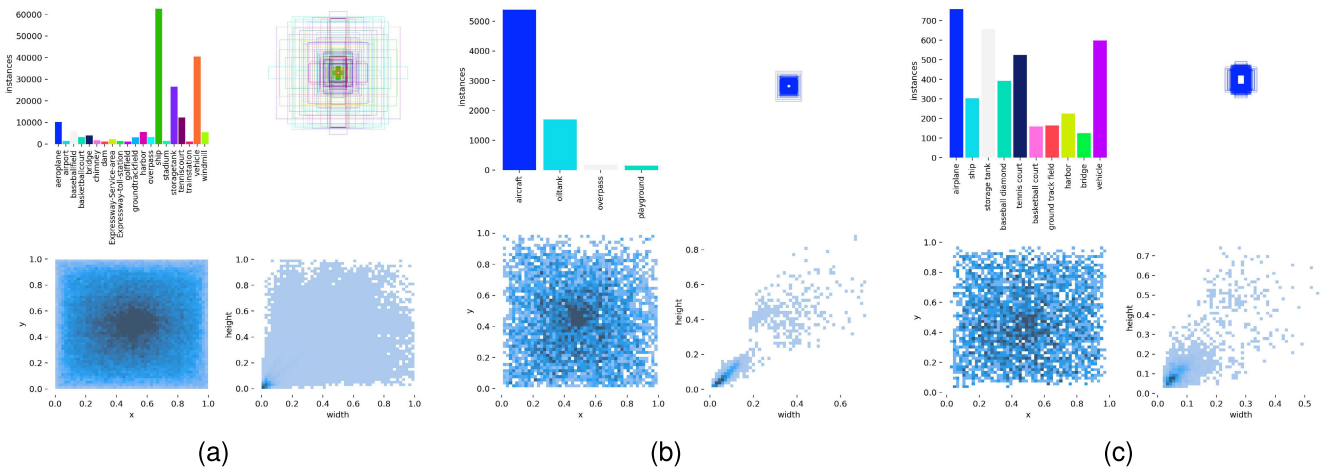


Fig. 6. Statistical characterization of instance quantities and size distributions across three benchmark datasets. (a)–(c) DIOR, RSOD, and NWPU VHR-10 datasets, respectively.

mAP measures overall performance across multiple classes. Following standard practice, results are reported at different intersection-over-union (IoU) thresholds: mAP_{50}, mAP_{75} and $mAP_{50\text{-}95}$. In this work, AP and mAP are served as our principal evaluation benchmarks. In addition, Precision, Recall, and $mAP_{50\text{-}95}$ are used as auxiliary metrics to measure the detection performance of the model. Specifically, TP is defined as the number of true positives, FN is defined as the number of false negatives, FP is defined as the number of false positives, and TN is defined as the number of true negatives. The specific formulas are as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{23}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{24}$$

$$\text{AP} = \int_0^1 P(R)\, dR \tag{25}$$

$$\text{mAP} = \frac{1}{n} \sum_{i=1}^{n} \int_0^1 P(R) dR. \tag{26}$$

C. Analysis of Experimental Datasets

Public datasets not only provide researchers with a comprehensive object detection benchmark but also offer rich data support for practical applications. In this article, the performance of the proposed method is tested and evaluated using three datasets: RSOD, NWPU VHR-10, and DIOR. The number of instances and size distribution in the three datasets are shown in Fig. 6.

1) DIOR: The detection in optical remote sensing images (DIOR) dataset [42] is a large-scale, multiclass, multiscale, high-resolution benchmark dataset for object detection in optical remote sensing imagery. It represents the largest publicly available benchmark of its kind. The dataset comprises 23 453 images containing 192 472 annotated instances, spanning 20 distinct object categories. These categories encompass vehicles, buildings, infrastructure, and natural objects. Each image has a fixed size of 800 × 800 pixels, with spatial resolutions ranging from 0.5 to 30 m/pixel. This diversity in scale and content makes the DIOR dataset well-suited for evaluating object detection tasks across diverse environments, including urban areas, rural landscapes, and industrial zones.

2) RSOD: The remote sensing object detection (RSOD) dataset [43] is a publicly available dataset specifically designed for object detection tasks in remote sensing imagery. It is widely adopted as an evaluation standard for benchmarking and comparing the performance of various remote sensing object detection algorithms, particularly in scenarios involving multiscale objects, cluttered backgrounds, and diverse object categories. The dataset comprises 976 images with 6950 annotated instances, organized into four distinct object categories.

3) NWPU VHR-10: The NWPU VHR-10 dataset [44] is a publicly available benchmark specifically designed for object detection tasks in remote sensing imagery. It comprises 800 high-resolution optical remote sensing images featuring ten distinct geo-spatial object categories, with 3651 meticulously annotated instances. The imagery exhibits spatial resolutions ranging from 0.5 to 2 m, providing high-quality visual data coupled with precise annotations. Characterized by its diverse object categories and varied scene complexity, this dataset serves as a significant benchmark for evaluating and advancing algorithms in the field of remote sensing object detection.

D. Comparison of Quantitative Results

To demonstrate the excellent detection performance, robustness, and generalization capabilities of the proposed method in remote sensing image object detection, we conducted extensive experiments on three datasets: DIOR, RSOD, and NWPU VHR-10. Since the aforementioned three datasets differ in terms of object dimensions, object category, and dataset scales, leading to differential experimental outcomes across benchmarks. We compared various detectors, including single-stage, two-stage, anchor-free, and Transformer-based detection networks. Furthermore, we provide detailed performance analyses of the proposed method on all three datasets, with quantitative results comprehensively reported in Tables II–IV.

1) Evaluation on DIOR: We evaluated the proposed DBCFS-Net method on the DIOR dataset and conducted

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TABLE II

QUANTITATIVE COMPARISON BETWEEN THE PROPOSED DBCFS-NET AND EXISTING APPROACHES ON THE DIOR DATASET. IN THIS TABLE, C1–C20 DENOTE THE 20 OBJECT CLASSES: AIRPLANE, AIRPORT, BASEBALL FIELD, BASKETBALL COURT, BRIDGE, CHIMNEY, DAM, EXPRESSWAY SERVICE AREA, EXPRESSWAY TOLL STATION, GOLF COURSE, GROUND TRACK FIELD, HARBOR, OVERPASS, SHIP, STADIUM, STORAGE TANK, TENNIS COURT, TRAIN STATION, VEHICLE, AND WINDMILL. THE RED FONT INDICATES THE OPTIMAL RESULT, WHILE THE BLUE FONT INDICATES THE SECOND-BEST RESULT

Methods	Backbone	mAP(%)	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
DETR [45]	ResNet50	68.6	39.6	74.0	65.2	80.7	26.5	75.2	66.8	70.5	52.5	74.2	62.1	27.4	47.0	8.5	46.0	14.5	64.4	54.3	14.3	55.6
TOOD [46]	ResNet101	79.9	93.9	91.4	94.0	85.0	55.9	85.5	72.3	90.0	79.6	84.9	88.0	59.1	67.9	72.1	93.4	73.4	94.6	62.0	61.6	93.4
TBNet [16]	ResNet50	73.6	64.9	86.8	76.6	89.2	50.6	80.0	74.3	86.4	74.7	85.2	83.6	56.4	64.9	79.1	77.8	58.1	88.2	69.4	44.6	87.0
MSNet [47]	CSPDarkNet53	75.3	95.3	67.4	91.0	90.2	44.6	82.6	49.2	78.4	64.8	73.4	85.1	66.4	61.8	94.4	90.4	84.5	94.8	48.1	58.9	85.4
RAST-YOLO [48]	CSPDarkNet53	69.8	84.3	76.4	78.7	85.9	40.2	76.8	50.2	62.6	56.5	77.1	73.7	61.1	56.6	91.1	74.3	77.9	89.3	53.3	54.0	76.2
CoF-Net [12]	ResNet50	75.8	84.0	85.3	82.6	90.0	47.1	80.7	73.3	89.3	74.0	84.5	83.2	57.4	62.2	82.9	77.6	68.2	89.9	68.7	49.3	85.2
TransMIN [49]	ResNet50	73.0	62.6	82.7	77.1	90.3	50.6	79.5	69.8	81.2	73.1	82.7	83.5	56.6	63.8	76.2	73.5	62.3	88.4	63.6	52.3	89.5
KLDet [50]	ResNet50	73.5	67.5	81.1	75.5	88.2	47.4	78.6	65.0	82.9	72.7	79.9	80.8	57.1	61.2	91.2	74.6	75.2	87.4	56.9	56.0	90.2
AFGMFNet [14]	DarkNet53	73.2	90.9	72.8	79.3	89.7	44.7	81.4	59.3	66.0	62.7	73.8	79.2	65.0	61.7	91.7	78.6	75.8	90.7	60.0	58.0	83.1
BSSOCNet [51]	CSPDarkNet53	70.7	87.4	78.4	74.5	89.1	44.3	77.6	59.8	58.5	60.2	78.2	71.8	62.4	59.2	89.7	65.7	74.0	88.5	55.6	54.4	84.8
CFIMNet [52]	ResNet50	71.2	72.5	74.1	79.0	89.8	45.6	79.1	67.7	82.3	77.6	80.7	81.6	38.4	59.5	75.9	67.4	68.6	89.9	51.8	52.7	89.6
DCDet [53]	ResNet152	78.3	90.1	77.6	88.2	90.6	58.0	85.9	61.1	85.7	80.6	83.8	87.7	55.9	68.4	91.3	79.8	69.4	91.7	72.9	65.2	79.5
SESANet [54]	DarkNet53	74.6	93.5	78.2	91.9	82.2	46.8	89.3	61.4	67.2	67.5	73.3	76.7	55.8	62.0	78.8	84.8	82.4	89.8	54.4	76.7	79.1
SFSANet [55]	ResNet50	74.6	85.8	81.3	85.8	84.7	46.8	76.5	66.2	83.0	74.5	71.2	76.6	60.9	61.6	88.2	81.4	77.4	81.8	59.5	65.9	82.7
SRAF-Net [56]	ResNet101	83.2	95.7	89.6	95.5	87.3	59.2	90.1	68.1	93.2	85.6	85.7	92.5	69.9	69.4	91.5	92.4	83.9	93.5	54.8	73.4	92.5
MSCF-CEM [57]	ResNet50	85.1	97.5	91.2	95.6	90.2	64.1	91.3	65.2	94.1	87.5	87.8	91.4	71.7	70.3	92.6	93.7	92.5	95.6	63.1	72.4	94.8
SCENE-YOLO [58]	CSPDarkNet53	87.1	98.4	93.2	97.0	91.8	66.2	92.4	66.9	95.9	88.4	90.1	92.9	75.2	71.4	96.5	96.2	93.1	98.6	64.7	75.1	97.6
Ours	CSPDarkNet53	87.8	97.0	95.7	95.8	93.0	61.7	89.6	89.4	97.4	83.2	93.5	91.7	77.0	74.9	94.1	98.8	87.5	97.2	83.9	65.1	91.9



Fig. 7. Exemplary prediction results obtained by the proposed DBCFS-Net framework on the DIOR dataset.

comparisons against state-of-the-art approaches, including DETR, TOOD, TBNet, MSNet, RAST-YOLO, CoF-Net, TransMIN, KLDet, AFGMFNet, BSSOCNet, CFIMNet, DCDet, SESANet, SFSANet, SRAF-Net, MSCF-CEM, and SCENE-YOLO. The comparative results, as summarized in Table II, demonstrate that DBCFS-Net gains mAP of 87.8%, and attains superior performance specifically on the airport, basketball court, dam, expressway service area, golf course, harbor, overpass, stadium, and train station. Among these categories, bridges and vehicles failed to achieve leading detection performance comparable to most other categories. We attribute this primarily to the fact that bridges typically exhibit extreme aspect ratios and linear geometric shapes, which inherently conflict with the default square anchor boxes or bounding box priors used by most detectors. Vehicles are typically among the smallest and most densely packed objects in datasets. Furthermore, the vehicle category exhibits extremely high appearance diversity (sedans, trucks, and construction vehicles) and is frequently occluded, significantly increasing the difficulty of learning. Representative detection outcomes are illustrated in Fig. 7.

2) Evaluation on RSOD: To validate the effectiveness of the proposed DBCFS-Net on the RSOD dataset, we conducted comparative experiments against state-of-the-art approaches,

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Fig. 8. Exemplary prediction results obtained by the proposed DBCFS-Net framework on the RSOD dataset.

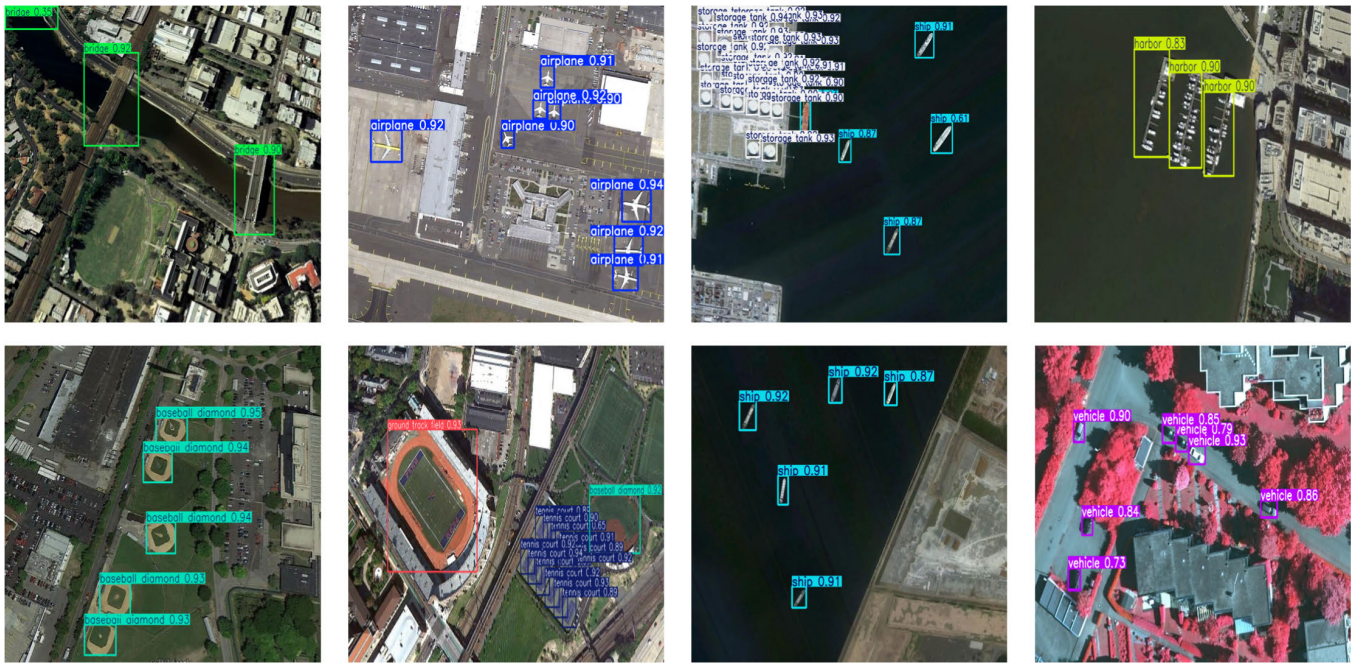


Fig. 9. Exemplary prediction results obtained by the proposed DBCFS-Net framework on the NWPU VHR-10 dataset.

TABLE III

QUANTITATIVE COMPARISON BETWEEN THE PROPOSED DBCFS-NET AND EXISTING METHODS ON THE RSOD DATASET IS CONDUCTED. THE RED FONT INDICATES THE OPTIMAL RESULT, WHILE THE BLUE FONT INDICATES THE SECOND-BEST RESULT

Methods	Backbone	mAP(%)	Aircraft	Oil Tank	Overpass	Playground
ABNet [11]	ResNet50	94.2	91.5	96.1	89.6	99.4
DA2FNet [59]	Hourglass	94.8	95.7	97.7	90.6	95.1
CANet [60]	ResNet101	95.2	91.8	97.1	94.1	97.9
CoF-Net [12]	ResNet50	95.0	96.5	98.8	85.5	99.3
SESANet [54]	DarkNet53	95.2	95.7	97.9	90.2	96.8
SFSANet [55]	ResNet50	95.0	96.0	97.1	88.4	98.7
MSNet [47]	CSPDarkNet53	95.6	96.4	97.3	90.9	98.0
SCENE-YOLO [58]	CSPDarkNet53	95.5	95.1	97.7	90.5	98.7
Ours	CSPDarkNet53	96.9	97.0	99.0	92.7	98.8

including ABNet, ATTS, DA2FNet, CANet, CoF-Net, SESANet, SFSANet, and MSNet. As presented in Table III, the results demonstrate that DBCFS-Net achieves the highest mAP of 96.9%, surpassing all compared methods. This represents performance gains of 1.3%–2.7% over other leading techniques. Furthermore, DBCFS-Net attains superior performance specifically on the aircraft, oil tank, and overpass object categories. Representative detection results are illustrated in Fig. 8.

3) Evaluation on NWPU VHR-10: To further validate the effectiveness of the proposed DBCFS-Net, we conducted comprehensive comparative experiments on the NWPU VHR-10 dataset against state-of-the-art approaches, including CANet, MEDNet, EGAT-LSTM, NL-LFPN-MR, CoF-Net, TMAFNet, ABNet, AGMFNet, GLSANet, SCNet, MSNet, LGDA, CFIMNet, TBNet, and BSSOCNet. The experimental results, detailed in Table IV, demonstrate that DBCFS-Net achieves the highest mAP of 96.6%. Furthermore, DBCFS-Net attains either the highest or second-highest precision among the compared methods specifically on the ship, storage tank, tennis court, and bridge object categories. Representative detection results are visually depicted in Fig. 9.

4) Discussion: The quantitative results presented in Tables II–IV demonstrate that DBCFS-Net achieves state-of-the-art or highly competitive performance across all three datasets. This can be directly attributed to the innovative design of our core architecture.

Thanks to the dynamic block downsampling module (DBDM), this module effectively mitigates the irreversible loss of high-frequency details—an inherent limitation of the fixed rigid downsampling strategy employed by baseline models. This enables DBCFS-Net to outperform models lacking such

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TABLE IV

A QUANTITATIVE COMPARISON BETWEEN THE PROPOSED DBCFS-NET AND EXISTING APPROACHES ON THE NWPU VHR-10 DATASET. IN THE TABLE, WHERE CATEGORIES C1-C10 DENOTE THE TEN OBJECT CLASSES: AIRPLANE, SHIP, STORAGE TANK, BASEBALL DIAMOND, TENNIS COURT, BASKETBALL COURT, ATHLETIC FIELD, HARBOR, BRIDGE, AND VEHICLE. THE RED FONT INDICATES THE OPTIMAL RESULT, WHILE THE BLUE FONT INDICATES THE SECOND-BEST RESULT

Methods	Backbone	mAP(%)	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
CANet [60]	ResNet101	93.3	99.9	85.9	99.3	97.3	97.8	84.8	98.4	90.4	89.2	90.3
MEDNet [61]	ResNet101	91.6	99.2	94.4	82.2	98.5	95.4	95.2	98.3	88.1	75.1	89.3
EGAT-LSTM [62]	GCN+LSTM	92.0	97.3	96.7	97.2	96.5	86.6	94.5	94.2	86.2	80.1	90.8
NL-LFPN-MR [63]	ResNet101	93.2	100	89.5	90.9	96.8	96.7	99.2	100	90.1	79.1	90.2
CoF-Net [12]	ResNet50	94.5	100	90.9	96.1	98.8	91.1	95.8	100	91.4	89.7	90.8
TMAFNet [64]	ResNet50	95.3	99.9	93.4	96.0	99.6	96.4	97.7	98.2	94.0	89.6	88.4
ABNet [11]	ResNet50	94.2	100	92.6	97.8	97.8	99.3	96.0	99.9	94.3	69.0	95.6
AFGMFNet [14]	CSPDarkNet53	95.2	99.9	94.2	95.5	98.4	99.7	96.3	99.3	96.9	84.6	87.3
GLSANet [65]	ResNet50	94.6	99.9	95.8	97.1	99.4	98.8	86.1	99.5	97.5	84.8	86.8
SCNet [66]	CSPDarkNet53	95.1	98.0	83.6	97.4	97.4	97.9	97.8	97.3	95.7	92.5	92.9
MSNet [47]	CSPDarkNet53	96.0	100	88.9	96.9	99.9	96.8	98.2	97.6	97.1	88.7	95.8
LGDA [67]	CSPDarkNet53	94.1	98.8	91.3	95.7	98.4	96.9	94.6	94.2	93.1	89.8	88.0
CFIMNet [52]	ResNet50	94.8	100	86.7	92.5	96.3	91.4	95.8	100	97.1	95.0	93.2
TBNet [16]	ResNet50	95.4	100	95.1	97.4	97.4	97.6	99.0	100	94.6	80.8	92.7
BSSOCNet [51]	CSPDarkNet53	94.2	99.9	85.9	89.7	97.3	96.6	97.3	100	97.7	85.4	90.8
SCENE-YOLO [58]	CSPDarkNet53	95.0	99.4	94.3	90.4	99.4	96.2	98.0	99.5	92.5	90.0	90.1
Ours	CSPDarkNet53	96.6	99.2	96.2	99.0	99.1	98.9	91.3	98.8	93.9	96.4	93.5

TABLE V

ABLATION EXPERIMENTS ON DIFFERENT DATASETS, WHERE A, B, AND C REPRESENT DBDM, CDAFEM, AND MB-LGFCPM

Method			mAP(%)		
A	B	C	DIOR	RSOD	NWPU VHR-10
			82.1	91.5	87.3
✓			83.2	92.6	90.9
	✓		84.6	94.4	92.7
		✓	84.1	93.6	92.5
✓	✓		86.9	94.9	93.5
✓		✓	86.2	95.4	94.2
	✓	✓	87.4	96.4	95.7
✓	✓	✓	87.8	96.9	96.6

mechanisms—such as MSNet, RAST-YOLO, and SCENE-YOLO—in fine-grained and small object detection tasks, as demonstrated on datasets like NWPU VHR-10. Compared with methods like CoF-Net and LEGNet that do not account for boundary information and overlook the importance of cross-domain collaborative coordination, DBCFS-Net achieves strong performance on the DIOR dataset characterized by complex object boundaries and cluttered backgrounds. We believe CDAFEM plays a crucial role in this achievement. The proposed CDAFEM significantly enhances the model's adaptability to scale variations and cluttered backgrounds while preserving feature discrimination integrity by imposing consistency constraints through cross-domain modeling. Compared with computationally intensive detectors such as DETR, TransMIN, and SESANet, DBCFS-Net demonstrates superior efficiency while achieving comparable or stronger global context capture capabilities. This objective is achieved through MB-LGFCPM's efficient establishment of long-range dependencies by integrating local detail features with global context features.

We objectively note that in certain specific categories or simple scenarios, some highly optimized baseline models may achieve comparable performance to ours, such as on the RSOD dataset. Our analysis suggests this may stem from the inherent simplicity of these scenarios or object categories, where the additional complexity of our model is unnecessary. In such cases, simpler architectures may approach the performance ceiling. This indirectly demonstrates the unique value of our model in complex scenarios.

E. Heat Map Visualization

To further visually demonstrate the enhancement effects of the DBDM, CDAFEM, and MB-LGFCPM on feature effectiveness representation and fusion within the network architecture, feature activation heat maps for airplanes, ships, storage tanks, and vehicles are visualized in Fig. 10. Fig. 10(b) demonstrates that after integrating the DBDM module, the activation intensity of object bodies is preserved while the background is attenuated. This observation indicates the critical role of DBDM in mitigating feature loss and alleviating feature information ambiguity. Fig. 10(c) reveals that upon CDAFEM incorporation, the heat map exhibits "block-like aggregation" with more uniform activation distribution, signifying enhanced model focus on object regions and improved adaptability to scale variations and complex backgrounds. Fig. 10(d) reveals graduated intensity transitions within the activation map that comprehensively cover object regions. This observation indicates that the integration of local detail

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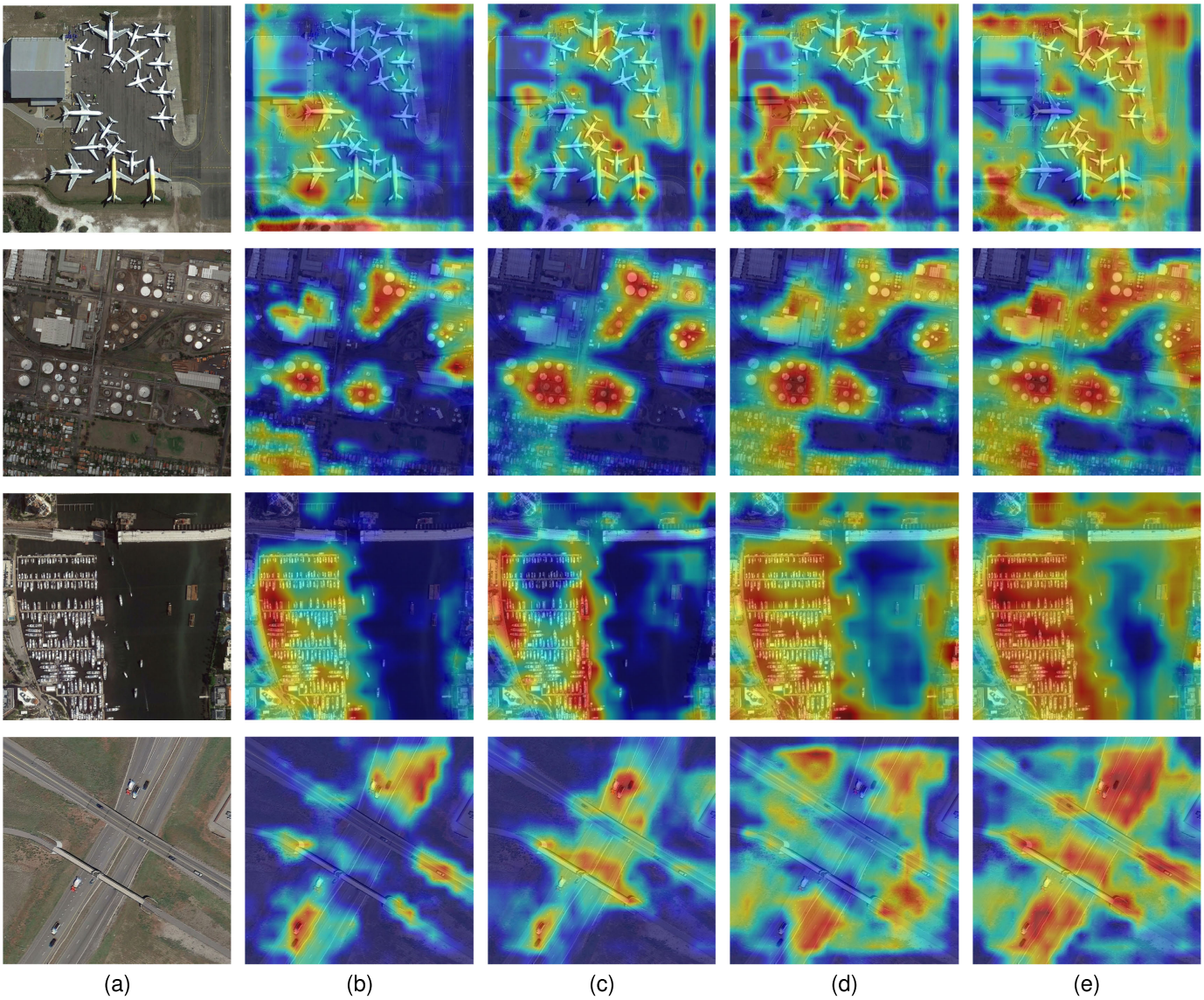


Fig. 10. Visual comparison of heat maps for different modules (a) Original Map. (b) +DBDModule. (c) +CDAFEModule. (d) +MB-LGFCPModule. (e) Our Method.

features with global contextual features strengthens detail perception while facilitating the global semantic relationship reasoning, establishing a synergistic complementary mechanism. Subsequently, as demonstrated in Fig. 10(e), discriminative feature locations are precisely covered with continuous and coherent object responses exhibiting precise spatial focusing. These visualizations collectively validate that the DBDN, CDAFEM, and MB-LGFCPM perform well in extracting effective features of the objects and eliminating local feature ambiguity and global feature blurring, playing a key role in improving the model's generalization and robustness.

F. Ablation Experiment

In this section, to validate the effectiveness of the modules proposed in this article, we designed a series of ablation experiments on the DIOR, RSOD, and NWPU VHR-10 datasets. Table V shows the quantitative data obtained by our proposed method under different metrics for different module combinations, further verifying the detection performance of the method proposed in this article.

To demonstrate the detection efficacy of DBCFS-Net on remote sensing imagery, we conduct comparative experiments on DIOR, RSOD, and NWPU VHR-10 datasets. As shown in Fig. 10, evaluations across Precision, Recall, and mAP50-95 metrics reveal consistent performance gains when incorporating individual components or their combinations. This upward trend confirms the positive contribution of each proposed component in DBCFS-Net. As depicted in Fig. 11, DBCFS-Net achieves Precision improvements of 4.3%, 5.2%, and 11.5% on the three datasets, respectively, demonstrating enhanced detection reliability and reduced false positives in remote sensing objects. The recall metrics show gains of 6.0%, 6.1%, and 9.4%, indicating improved object coverage with fewer missed detections. Furthermore, $mAP_{50\text{-}95}$ increases by 7.6%, 8.7%, and 16.9%, reflecting comprehensive advancements in localization accuracy and robustness. These consolidated

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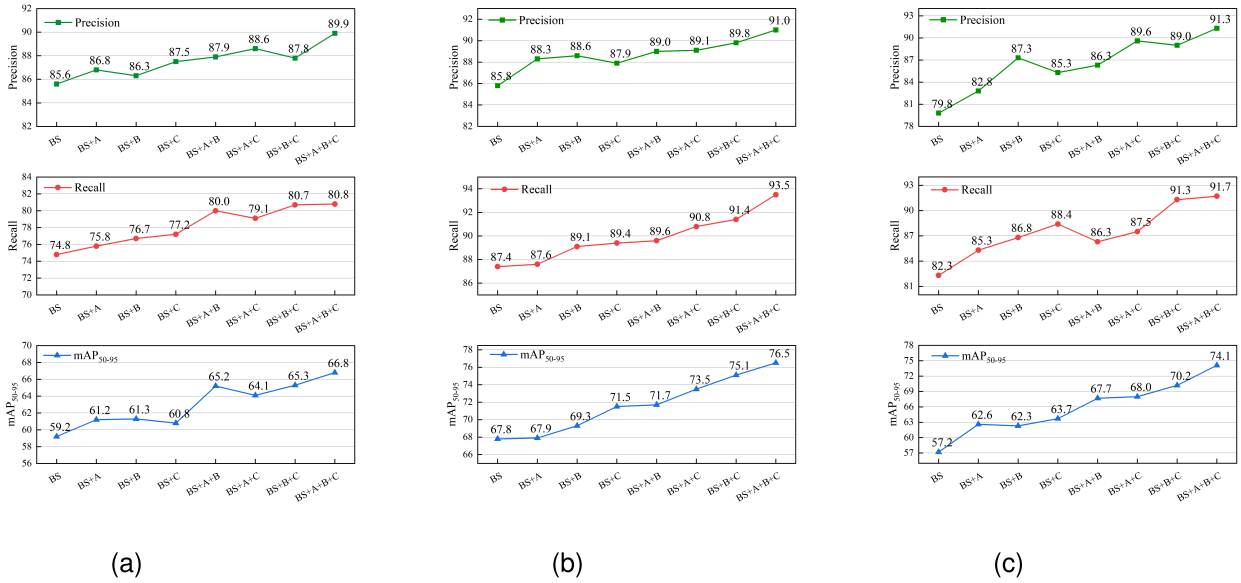


Fig. 11. Performance of DBCFS-Net across precision, recall, and $mAP_{50–95}$ metrics on different datasets. In the notation, BS denotes the baseline model, while A–C represent the DBDM, CDAFEM, and MB-LGFCPM components, respectively. (a) Precision, recall, and $mAP_{50–95}$ on DIOR. (b) Precision, recall, and $mAP_{50–95}$ on RSOD. (c) Precision, recal,l and $mAP_{50–95}$ on NWPU VHR-10.

improvements confirm the method's superior generalization capability across diverse operational environments.

1) Effect of DBDM: To validate the effectiveness of the DBDM, we conducted ablation studies comparing models with and without this module, as shown in Table V. The incorporation of this module contributes to enhancing the detection performance of the model. The DBDM operates by partitioning the feature map into uniformly sized subblocks that are processed individually. Each subblock can dynamically adapt based on the input content, thereby enhancing the model's adaptability to spatial variations. By establishing a dynamic mapping mechanism between the feature space and the semantic space, it effectively mitigates the limitations of insufficient feature representation and feature loss inherent in fixed-domain sampling schemes. The data in the table shows that, compared with the baseline model, performance improvements of varying degrees were achieved across different benchmark datasets. As can be seen from the data in the table, there are varying degrees of improvement across different datasets compared with the baseline model. In addition, as shown in the auxiliary metrics in Fig. 11, the inclusion of this module effectively improves detection performance, specifically: an increase of 1.2%, 2.5%, and 3.0% in the Precision metric; an increase of 1.0%, 0.2%, and 3.0% in the recall metric; and an increase of 2.0%, 0.1%, and 5.4% in the $mAP_{50–95}$ metric.

2) Effect of CDAFEM: The CDAFEM, proposed in DBCFS-Net, aims to establish consistency constraints through cross-domain joint modeling. This approach enhances the model's adaptability to scale variations and complex backgrounds while preserving discriminative feature information. Concurrently, it refines object boundaries and enhances edge structures by leveraging textural information, thereby reinforcing edge continuity. This mechanism provides a novel approach for improving the model's generalization capability and robustness. The tabular data presented in Table V demonstrate that performance surpasses that of the baseline model following the integration of CDAFEM. Also, the specific performance of the auxiliary indicators shown in Fig. 11 is as follows: the precision indicator improved by 0.7%, 2.8% and 7.5%; the Recall indicator improved by 1.9%, 1.7% and 4.5%; and the mAP_{50-95} indicator improved by 2.1%, 1.5% and 5.1%, respectively. This strongly proves the effectiveness of this module.

3) Effect of MB-LGFCPM: To address the discrepancies in object scales and the inherent conflict between local feature ambiguity and global feature indistinctness, we introduce the MB-LGFCPM. This module guides the model to effectively couple local details with global contextual information, thereby constructing a "detail-semantic" bidirectional reinforcement mechanism for feature representation. This mechanism facilitates the dynamic interaction and synergistic complementarity optimization among features. As shown in the data from Table V, the incorporation of MB-LGFCPM effectively improves detection performance, with a significant increase in the mAP metric compared with the baseline model. The module demonstrates significant performance gains across auxiliary metrics. The specific improvements are as follows: the precision indicator improved by 1.9%, 2.1%, and 5.5%; the Recall indicator improved by 2.4%, 2.0%, and 6.1%; and the mAP_{50-95} indicator improved by 1.6%, 3.7%, and 6.5%. These consistent enhancements validate the module's efficacy in comprehensive detection performance.

4) Analysis of Computational Cost and Inference Time: To further demonstrate that the proposed method also exhibits excellent real-time performance, we conducted ablation experiments on each component to evaluate computational complexity and real-time capabilities. The specific results are shown in Table VI. Compared with the baseline, we sacrificed computational complexity and detection speed in pursuit of

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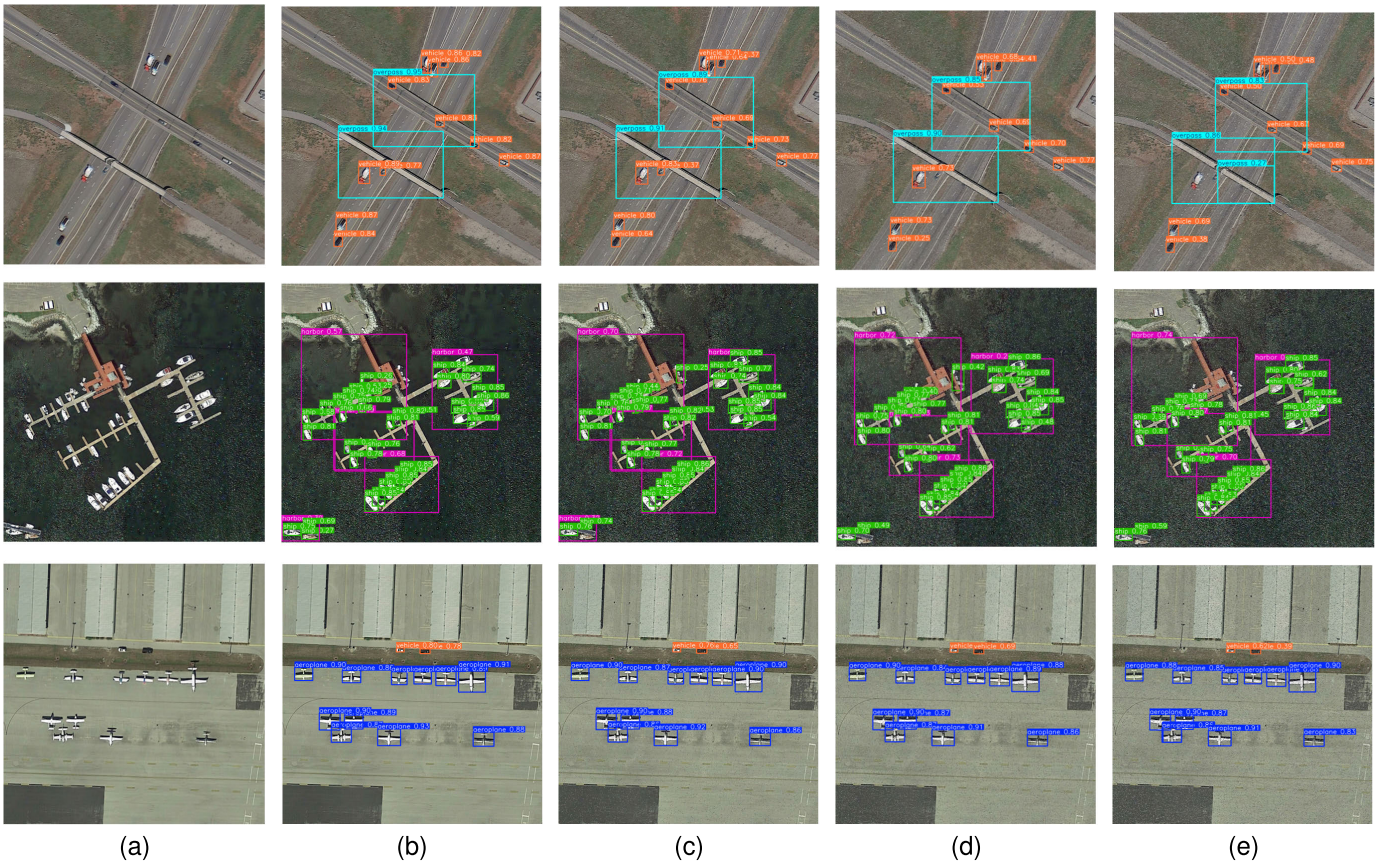


Fig. 12. Visualization of robustness and generalization capability. Comparison between original image and images with added noise of varying intensities (a) Original images. (b) Detection results. (c) Detection results with 5% noise. (d) Detection results with 10% noise. (e) Detection results with 15% noise.

TABLE VI
ABLATION STUDY ABOUT COMPUTATIONAL COMPLEXITY AND INFERENCE TIME

Method			Param(M)	FLOPs(G)	Inference_time(ms)	FPS
A	B	C				
			2.7	8.3	3.8	263.2
✓			5.7	13.1	6.2	161.3
	✓		18.8	25.7	12.4	80.6
		✓	4	8.6	5.7	175.4
✓	✓		21.6	29.6	12.6	79.4
✓		✓	7.1	13.5	6.8	147.1
	✓	✓	20	21	12.1	82.6
✓	✓	✓	23.1	25.9	15.4	64.9

higher detection accuracy. Among the three components we proposed, DBDM and MB-LGFCPM do not impose additional computational burden on the model. Considering their detection accuracy, these two components demonstrate significant advantages. In our proposed method, DBCFS-Net, the primary computational overhead stems from CDAFEM, which also contributes the greatest improvement in accuracy. In this ablation study, the computational overhead of DBCFS-Net reached its peak when combined with DBDM and CDAFEM, but this situation was alleviated after incorporating MB-LGFCPM. At the same time, we also compared our approach with the outstanding methods developed by other researchers in recent years, as detailed in Table VII. Meanwhile, we validate the effect of different types of wavelet transforms on the model on the RSOD dataset, details of which are shown in Table VIII. Haar wavelets are in the lead, both in terms of the number of covariates and the detection metrics.

As shown in Tables VI and VII, although our method exhibits certain improvements over the baseline in terms of parameter count, computational complexity, and inference time, it still maintains a leading position compared with the other methods listed. It ranks third in both parameter count and computational complexity, while achieving the fastest detection speed.

5) Analysis of the Robustness and Generalization Capability: It is well known that image degradation can significantly impact the detection performance of remote sensing images. Therefore, we employed images with added noise at three intensity levels—5%, 10%, and 15%—to test the model's robustness and generalization capability. In this experiment, we focused on the effects of object distribution and image degradation severity on detection performance. Comparative experimental results are shown in Fig. 12. As shown in the figure, the distribution of object locations in the original image has almost no effect on the model's object detection performance. However, in degraded images, we observe that the model's detection performance is affected to varying degrees. When noise intensity is increased by 5%, the model's detection performance shows no significant change, with no missed

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TABLE VII
COMPARISON WITH OTHER METHODS IN COMPUTATIONAL COMPLEXITY AND INFERENCE TIME

Method	Param(M)	FLOPs(G)	Inference_time(ms)	FPS
DETR [45]	49.2	95	50	20
TBNet [16]	58.7	-	41.7	24
MSNet [47]	29.8	98.1	47.2	21.2
CoF-Net [12]	39.4	-	55.6	18
TransMIN [49]	46.5	-	100	10
KLDet [50]	32.0	339.3	55	18.2
CFIMNet [52]	37.5	-	83.3	12
DCDet [53]	49.9	7.5	114.9	8.7
BSSOCNet [51]	19.4	26.1	76.9	13
SFSANet [55]	9.1	24.3	21.7	46
Ours	23.1	25.9	15.4	64.9

TABLE VIII
THE IMPACT OF DIFFERENT WAVELET TRANSFORM METHODS ON MODEL PARAMETERS

Wavelet Type	Haar	Daubechies	Symlets	Biorthogonal	Coiflets
Param/M	23.11	24.11	23.35	23.67	23.67
mAP(%)	96.9	96.1	96.3	96.4	95.4
mAP_{50-95}(%)	76.5	76.3	75.6	75.7	73.5
Precision(%)	91.0	91.1	91.1	90.1	90.6
Recall(%)	93.5	91.2	92.9	93.3	91.3

TABLE IX
COMPARATIVE EXPERIMENTS WITH TRAINING SETS OF DIFFERENT PROPORTIONS IN RSOD DATASET

Training Set Ratio	mAP(%)
70%	96.9%
60%	94.2%
50%	88.8%
40%	84.8%

detections or false positives occurring. When noise intensity is increased by 10%, missed detections of occluded objects begin to appear. When noise intensity is increased by 15%, missed detections also occur for densely distributed objects. Due to destructive interference in image quality, certain pixels of the object were damaged, ultimately leading to missed detections.

We also conducted comparative experiments on the RSOD dataset using training sets of varying proportions. We set up four control experiments in increments of 10%, with the specific results shown in Table IX.

As the proportion of the training set decreases, the detection accuracy of DBCSF-Net on the RSOD dataset declines. Even when the training set comprises only 40% of the data, the MAP value remains at 84.8%, which is still a relatively impressive detection accuracy. This further demonstrates the robust performance and strong generalization capabilities of our method.

V. CONCLUSION

In this article, we propose a novel method called DBCFS-Net to improve the detection capability of remote sensing images. This method consists of three modules: DBDM, CDAFEM, and MB-LGFCPM. It not only effectively alleviates the loss of high-frequency fine-grained features during sampling, but also captures efficient and discriminative object features through multiscale modeling. In addition, establishing long-range dependencies based on feature similarity to form a local–global feature synergy mechanism plays a crucial role in improving the separability between the background and the objects. Extensive experiments conducted on the DIOR, RSOD, and NWPU VHR-10 benchmark datasets demonstrate that DBCFS-Net achieves superior detection performance, outperforming current state-of-the-art detectors. Specifically, on these three datasets, DBCFS-Net achieved mAP scores of 87.8%, 96.9%, and 96.6%, respectively. However, our comprehensive experimental analysis shows that the continuous pursuit of higher detection accuracy inevitably leads to an increase in the number of parameters, which results in a lack of overall lightweight design of the network architecture and an increase in the training cost. As an important part of our future work, we will carry out research on model lightweighting techniques based on branch reduction, quantization, and knowledge distillation, with a view to significantly reducing the model size and improving the inference speed, while maintaining its high accuracy advantage as much as possible. We will also explore the possibilities of its deployment in edge devices, such as utilizing inference engines like TensorRT and ONNX Runtime, to achieve extreme acceleration on platforms like NVIDIA Jetson and FPGAs. At the same time, we have shortcomings when dealing with the task of detecting objects with extreme aspect ratios, large geometries, and small sizes. The task of directional object detection also faces limitations. Future research will improve the adaptability to linear and small objects from the perspective of network detection head, bounding box representation, and optimization of loss function and matching strategy, as well as expand the capability of directional object detection. Also, this study has not yet explored the potential advantages of integrating multisource data to enhance detection performance; effective utilization of multimodal information will also be a key direction for subsequent research.

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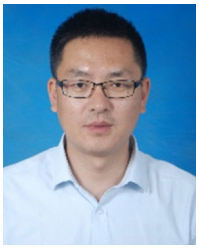
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